Learning Objective 7.3: I can use Ampere's Law to calculate the magnetic field at an observation point for simple magnetostatic systems.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Circulation and enclosed current.** Three long straight wires run perpendicular to the page. Wire 1 carries 5 A out of the page, wire 2 carries 3 A into the page, and wire 3 carries 2 A out of the page. A closed path C in the page encloses wires 1 and 2 but not wire 3.

   (a) Find $\oint_C \mathbf{H}\cdot d\mathbf{l}$ when we traverse C counterclockwise.

   Counterclockwise travel makes out of the page the positive direction by the right-hand rule. Wire 3 is outside and contributes nothing:

   $$\oint_C \mathbf{H}\cdot d\mathbf{l} = I_{\text{encl}} = 5 - 3 = 2 \text{ A}$$

   circulation = 2 A

   (b) A second path encloses all three wires and we traverse it clockwise. Find its circulation.

   Clockwise travel makes into the page the positive direction:

   $$\oint \mathbf{H}\cdot d\mathbf{l} = -5 + 3 - 2 = -4 \text{ A}$$

   circulation = -4 A

   (c) Ampere's law gives the circulation around C exactly. Explain why it does not give the field **H** at a point on C for this arrangement.

   Ampere's law fixes only the integral of $\mathbf{H}\cdot d\mathbf{l}$ around the whole path. To find H at a point, H must come out of the integral, which requires a path on which **H** is tangent and constant. Three wires at different positions have no such symmetry, so we add the three line-current fields by superposition instead.

2. **A round copper wire.** A long copper wire of radius $a = 1$ mm carries a steady $I = 5$ A, spread uniformly over its cross section.

(a) Use Ampere's law to derive $H_\phi(\rho)$ inside the wire.

By symmetry $\mathbf{H} = \hat{\phi}\, H_\phi(\rho)$, and the path is a circle of radius $\rho < a$. The current density is $J = I/(\pi a^2)$:

$$\oint_C \mathbf{H}\cdot d\mathbf{l} = 2\pi\rho\, H_\phi$$

$$I_{\text{encl}} = \int_0^{2\pi}\int_0^{\rho} J\,\rho'\,d\rho'\,d\phi = J\pi\rho^2 = I\frac{\rho^2}{a^2}$$

$$H_\phi = \frac{I\rho}{2\pi a^2}$$

(b) Find H_ϕ at $\rho = 0.5$ mm, 1 mm, and 2 mm.

$$H(0.5\text{ mm}) = \frac{5\times 0.5\times 10^{-3}}{2\pi\times(10^{-3})^2} = 398\text{ A/m}$$

$$H(1\text{ mm}) = \frac{5}{2\pi\times 10^{-3}} = 796\text{ A/m}$$

$$H(2\text{ mm}) = \frac{5}{2\pi\times 2\times 10^{-3}} = 398\text{ A/m}$$

0.5 mm: 398 A/m

1 mm: 796 A/m

2 mm: 398 A/m

(c) Find B at the surface of the wire.

$$B = \mu_0 H = 4\pi\times 10^{-7}\times 796 = 1.00\text{ mT}$$

$B(a)$ = 1.00 mT

(d) Use the point form $\nabla\times\mathbf{H} = \mathbf{J}$ to recover the current density from your answer to part (a), and check it against $I/(\pi a^2)$.

Only the z component of the curl survives for $\mathbf{H} = \hat{\phi}\, H_\phi(\rho)$:

$$J_z = \frac{1}{\rho}\frac{d}{d\rho}(\rho H_\phi) = \frac{1}{\rho}\frac{d}{d\rho}\left(\frac{I\rho^2}{2\pi a^2}\right) = \frac{I}{\pi a^2}$$

$$= \frac{5}{\pi\times(10^{-3})^2} = 1.59\times 10^6\text{ A/m}^2$$

3. **The point form with a nonuniform current.** Inside a long wire of radius $a = 2$ mm, the measured field is $\mathbf{H} = \hat{\phi}\, H_0\,(\rho/a)^2$ with $H_0 = 100$ A/m.

   (a) Find the current density $\mathbf{J}(\rho)$ inside the wire.

$$J_z = \frac{1}{\rho}\frac{d}{d\rho}\left(\rho\,\frac{H_0\rho^2}{a^2}\right) = \frac{1}{\rho}\,\frac{3H_0\rho^2}{a^2} = \frac{3H_0\rho}{a^2}$$

   The current is densest at the surface and zero on the axis.

   (b) Find the total current in the wire two ways: from Ampere's law on a circle of radius a, and by integrating $\mathbf{J}$ over the cross section.

I = 1.26 A

$$I = \oint \mathbf{H}\cdot d\mathbf{l} = 2\pi a H_0 = 2\pi \times 0.002 \times 100 = 1.26\text{ A}$$

$$I = \int_0^{2\pi}\int_0^a \frac{3H_0\rho}{a^2}\,\rho\,d\rho\,d\phi = \frac{3H_0}{a^2}\,\frac{a^3}{3}\,2\pi = 2\pi a H_0$$

   The two agree, as Stokes' theorem requires.

   (c) Find J at the surface of the wire.

$$J(a) = \frac{3H_0}{a} = \frac{3\times 100}{0.002} = 1.5\times 10^5\text{ A/m}^2$$

$J(a)$ = 1.5×10^5 A/m^2

4. **Two parallel wires.** Two long parallel wires 2 cm apart carry 10 A in opposite directions, as in a two-wire line.

   (a) Find H midway between the wires.

   Each wire gives $I/(2\pi\rho)$ at $\rho = 1$ cm, and between opposite currents the two fields point the same way, so they add:

$$H = 2\times\frac{10}{2\pi\times 0.01} = 318\text{ A/m}$$

H = 318 A/m

   (b) Find H on the line through both wires, 1 cm outside one of them.

   The point is 1 cm from the near wire and 3 cm from the far one, and outside the pair the two fields point in opposite directions:

$$H = \frac{10}{2\pi\times 0.01} - \frac{10}{2\pi\times 0.03} = 159 - 53 = 106\text{ A/m}$$

H = 106 A/m

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